Exploring Timbre Spaces through Dimensionality Reduction and Clustering

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***Abstract*—This work investigates the relationship between perceptual and computational timbre spaces by applying dimensionality reduction techniques to audio features extracted from musical recordings. A subset of three stylistically related genres —Samba, Jazz, and Afrobeat—was selected, and a comprehensive set of features, including harmonic and percussive descriptors, was extracted. Dimensionality reduction methods such as PCA, t-SNE, and UMAP were used to visualize the resulting timbre space. Clustering algorithms (K-Means and HDBSCAN) were applied to evaluate the consistency of the computed space with genre and cultural groupings. Interestingly, instances of perceptual similarity—such as João Gilberto's "Desafinado" aligning with Jazz clusters—suggest that the computational space is capable of capturing musically meaningful relationships. The results indicate that appropriate feature design and dimensionality reduction techniques can, to some extent, reflect human-perceived timbral similarities across styles and cultural contexts.**

***Index Terms*—Timbre spaces, Dimensionality reduction, Clustering, Musical analysis, Data visualization**

I. Introduction

Timbre is the perceptual attribute that allows listeners to distinguish between sounds that share the same pitch, loudness, and duration. The American National Standards Institute (1973) defines it as the quality of sound that differentiates two tones with identical pitch and loudness, from distinct sources. Despite its central role in auditory perception and musical expression, timbre remains one of the most complex features to define and quantify.

Early studies in psychoacoustics and cognitive psychology (Grey, 1977; Von Bismarck, 1974) showed that timbre perception is multidimensional, involving descriptors such as brightness, warmth, or roughness. These perceptual dimensions were further explored using listener surveys and verbal reports (Pratt & Doak, 1976; Zacharakis & Reiss, 2011), revealing consistent patterns in how people describe and group different timbres.

In parallel, computational approaches have emerged to represent timbre by extracting features from audio signals, aiming to build "timbre spaces" using dimensionality reduction and clustering algorithms (McAdams & Bigand, 1993; Sethares, 2005). However, the correspondence between these computational representations and perceptual similarity remains an open question.

This study investigates the relationship between perceptual and computational timbre spaces by focusing on three stylistically and rhythmically related musical genres: Samba, Jazz, and Afrobeat. These genres, although culturally distinct, share instrumental and structural characteristics that may produce overlapping timbral features.

By extracting harmonic and percussive descriptors from audio recordings and applying dimensionality reduction techniques (PCA, t-SNE, UMAP), followed by clustering algorithms (K-Means, HDBSCAN), we evaluate how well the computational timbre space aligns with genre distinctions and perceptual similarities. In particular, we explore whether such spaces can capture musical relationships that go beyond simple stylistic labels.

A. Related Work

The study of musical timbre has historically drawn from both perceptual and acoustic perspectives. Foundational works such as Grey (1977) and von Bismarck (1974) used multidimensional scaling and verbal descriptors to map how listeners perceive timbral differences, establishing the notion of a "timbre space" based on human judgment. Further studies (e.g., McAdams & Bigand, 1993; Pratt & Doak, 1976) contributed to the understanding of cognitive and subjective dimensions of timbre, emphasizing its multidimensional nature.

On the computational side, authors like Sethares (2005) and Howard & Angus (2017) explored how spectral and temporal features can be used to characterize timbre in measurable terms. These works highlight the technical challenge of bridging physical signal descriptors with the richness of perceptual experience.

More recent efforts have aimed to combine these two views. Ganguli et al. (2020), for example, mapped the timbral landscape of Arabic music by using Harmonic-Percussive Source Separation (HPSS) and a large set of audio features. Dimensionality reduction techniques were applied to visualize stylistic similarities and cultural relationships, indicating that computational models can reflect aspects of timbral perception.

However, few studies attempt a direct comparison between perceptual expectations and the structure of computationally derived timbre spaces. Additionally, the use of clustering metrics as a way to evaluate the coherence of these spaces—particularly in terms of genre or cultural grouping—remains underexplored. This work seeks to contribute to that gap by combining perceptually informed genre selection with quantitative embedding analysis and unsupervised clustering.

II. Methods

This section describes the methodology adopted to construct and analyze the computational timbre space. The process includes dataset construction, feature extraction, dimensionality reduction, and cluster evaluation. Each step is designed to investigate whether perceptually similar musical pieces are also grouped closely in the computed space.

A. Dataset

In this project, both quantitative and qualitative evaluations were considered. To enable a more perceptual analysis, a limited and well-curated set of songs was selected. The underlying assumption is that meaningful evaluation requires familiarity with the material—knowing the songs and recognizing which ones sound perceptually similar. For this reason, Afrobeat, Jazz, and Samba was selected as the target genres. Although it may be difficult to articulate precisely what makes them similar, these genres are often perceived as *timbrally* related. This subjective perception provides an interesting scenario to explore the correspondence between perceptual and computational timbre spaces.

Each track was manually tagged with its corresponding musical style and country of origin. Since each style is uniquely associated with a specific country in this dataset (e.g., Jazz from the USA, Samba from Brazil, Afrobeat from Nigeria), these two labels are treated as interchangeable for the purposes of evaluation.

All files were converted to mono WAV format and trimmed to a fixed duration of 200 seconds to reduce variability due to track length.

B. Feature Extraction

To extract the relevant temporal and spectral features of each audio recording, the *Librosa Library* was used. The audio duration was fixed to 200 seconds, and resampled to 22050 to preserve audio fidelity and capture frequencies lower than 11kHz, which is appropriate for musical analysis. A technique called Harmonic-Percussive Source Separation (HPSS) was used to enable the extraction of specialized features from each component.

Global features such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroids, spectral roll-off, and spectral bandwidth were computed directly from the full audio signal. From the harmonic component, chroma and tonnetz features were extracted to capture pitch content and tonal relationships. From the percussive component, zero-crossing rate (ZCR), spectral flatness, and root mean square (RMS) energy were computed to capture rhythmic and noise-like properties.

Each feature was aggregated by computing the mean and standard deviation, resulting in a feature vector for each song. This feature extraction process generated a total of 74 features.

C. Feature Normalization

Before applying dimensionality reduction and clustering algorithms, all audio features were normalized using z-score normalization. This process transforms each feature to have zero mean and unit variance, preventing features with larger numerical ranges from dominating the results. The normalization was performed using the `StandardScaler` method from the scikit-learn library.

TABLE I: Hyperparameters used for Clustering

Method	Parameters
K-means	n_cluters = 3
HDBSCAN	min_cluster_size = 8

TABLE II: Hyperparameters used for dimensionality reduction

Method	Parameters
PCA	—
t-SNE	perplexity = 5, random_state = 42
UMAP	random_state = 42

TABLE III: Manually Created Dataset

Style	Songs
Afrobeat	30
Jazz	30
Samba	30

D. Dimensionality Reduction

To investigate the timbre space visually, three dimensionality reduction techniques were applied to the normalized features: Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP). These techniques transform the 74-dimensional feature vectors to two dimensions, allowing a representation that captures the most relevant similarities between songs.

While PCA provides a linear projection based on variance maximization, t-SNE and UMAP offer nonlinear mappings that better preserve local relationships and clustering tendencies within the data.

E. Clustering and Evaluation

After projecting the songs into a two-dimensional space, clustering algorithms were applied to investigate the grouping behavior within the timbre space. Two unsupervised clustering techniques were used: K-Means and HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise).

K-Means was configured to form three clusters, reflecting the number of genres in the dataset. This method assigns each point to one of the clusters by minimizing the distance to the cluster centroids.

HDBSCAN, on the other hand, does not require the number of clusters to be defined in advance. Instead, it identifies clusters of varying densities and treats less consistent points as outliers. This flexibility makes HDBSCAN particularly appropriate for exploring the structure of the timbre space, especially in the presence of overlapping genre characteristics.

Both clustering methods were applied over the 2D representations produced by PCA, t-SNE, and UMAP. The quality of the resulting clusters was evaluated using internal metrics such as Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index.

TABLE IV: Extracted Audio Features

Feature	Type	Description
MFCC1–13	Global	Mel-Frequency Cepstral Coefficients; capture timbral envelope
Spectral Centroid	Global	Indicates the “brightness” of the sound
Spectral Rolloff	Global	Frequency below which a percentage of total energy lies
Spectral Bandwidth	Global	Width of the frequency band in the spectrum
Chroma 1–12	Harmonic	Energy distribution across pitch classes (C to B)
Tonnetz 1–6	Harmonic	Tonal relations in pitch space (tonal centroid features)
ZCR	Percussive	Zero-Crossing Rate; indicates noisiness/rhythm
Flatness	Percussive	Spectral flatness; relates to tonality vs noise
RMS	Percussive	Root Mean Square energy; indicates signal amplitude

III. Results and Discussion

The results are divided into two main parts: a quantitative analysis and a qualitative analysis, which are explored in the following subsections. The goal is to validate whether numerical results can reflect human musical perception—that is, whether the computational space aligns with the perceptual space.

A. Overview of the Projections

To explore the structure of the timbre space, the 74-dimensional feature vectors were projected into two-dimensional space using PCA, t-SNE, and UMAP. Each projection provided a distinct view of the data distribution, allowing for both visual and numerical assessment of groupings.

The PCA projection showed limited separation between styles, forming a single dense cluster with no clear boundaries. This was expected, as PCA is a linear method and may not effectively preserve local relationships in complex, perceptually grounded data like timbre.

In contrast, t-SNE revealed more localized groupings. While some overlap between styles occurred—which is expected, given that the dataset was intentionally composed of perceptually similar genres—small clusters appeared, particularly around Jazz recordings.

The UMAP projection provided the most visually consistent structure, with two better defined clusters and one more diffuse group. One cluster was clearly associated with Jazz recordings, aligning with perceptual expectations.

To quantitatively evaluate the effectiveness of each projection, three unsupervised clustering metrics were computed using the known style labels: Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. As shown in Table IV, UMAP achieved the best overall performance across all three metrics, reinforcing the visual impression of better-defined stylistic structure.

TABLE V: Quantitative Evaluation of Dimensionality Reduction Techniques

Method	Silhouette	Calinski-Harabasz	Davies-Bouldin
PCA	0.050	16.537	3.429
t-SNE	0.064	9.440	2.715
UMAP	0.138	22.736	2.085

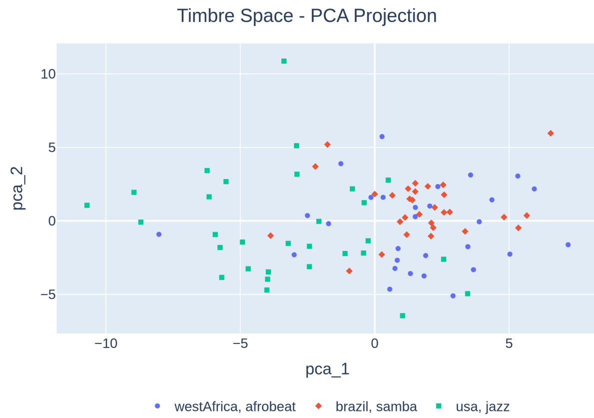

Fig. 1: Timbre Space - PCA

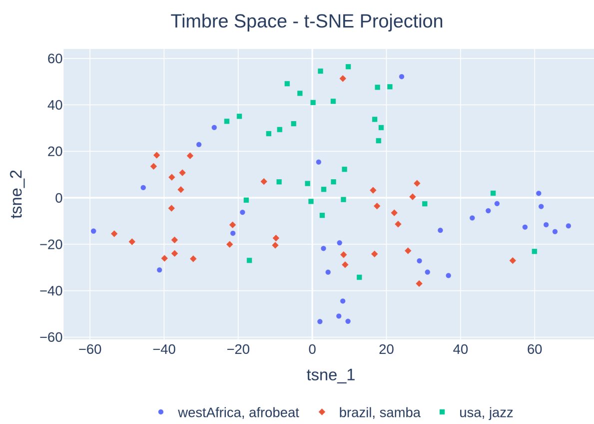

Fig. 2: Timbre Space - t-SNE.

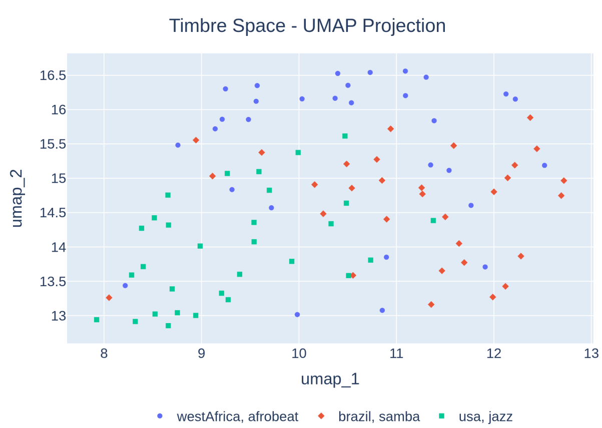

Fig. 3: Timbre Space - UMAP.

B. *Clustering*

As discussed in the previous sections, two clustering techniques were applied to the low-dimensional representations of the musical samples—K-means and HDBSCAN. Each combination of projection and clustering method was evaluated using the same validation metrics as before: Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. In this context, these metrics provide insight into the spatial arrangement and quality of the resulting clusters.

However, it is important to note that no evaluation metrics were reported for the combinations of HDBSCAN with PCA or t-SNE. In these cases, HDBSCAN was unable to form valid clusters, assigning all or nearly all data points to noise (i.e., the label -1). Since the clustering metrics rely on the existence of multiple distinct groups, the absence of valid clusters makes quantitative evaluation meaningless for those configurations.

Among the clustering results, UMAP combined with HDBSCAN presented the most coherent structure, achieving the highest Silhouette Score (0.535), the highest Calinski-Harabasz Index (55.737), and the lowest Davies-Bouldin Index (0.529). These values suggest compact, well-separated clusters that are consistent with the data's intrinsic structure.

In contrast, PCA followed by K-Means yielded the weakest results, particularly a Davies-Bouldin Index of 29.412, indicating poorly defined clusters with high overlap. This aligns with previous observations that PCA fails to expose meaningful separation in the data.

The t-SNE + K-Means and UMAP + K-Means combinations showed intermediate performance, with UMAP again outperforming t-SNE in all metrics. This supports the idea that UMAP is more effective at preserving both local and global structures in the reduced space.

TABLE VI: Clustering Evaluation Metrics

Method	Silhouette	Calinski-Harabasz	Davies-Bouldin
PCA + K-Means	0.095	31.722	29.412
t-SNE + K-Means	0.195	33.381	1.393
UMAP + K-Means	0.260	46.761	1.195
UMAP + HDBSCAN	0.535	55.737	0.529

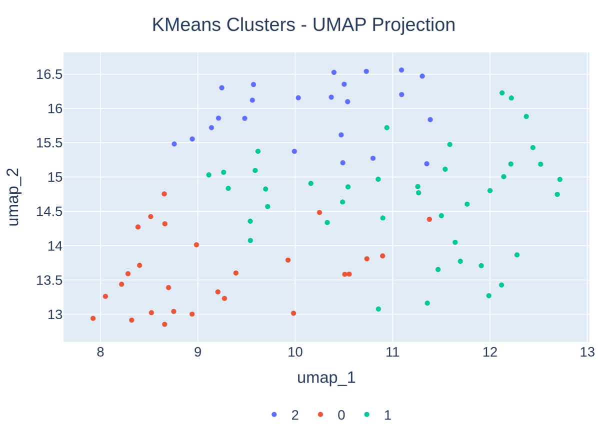

Fig. 4: K-Means and UMAP Clusters

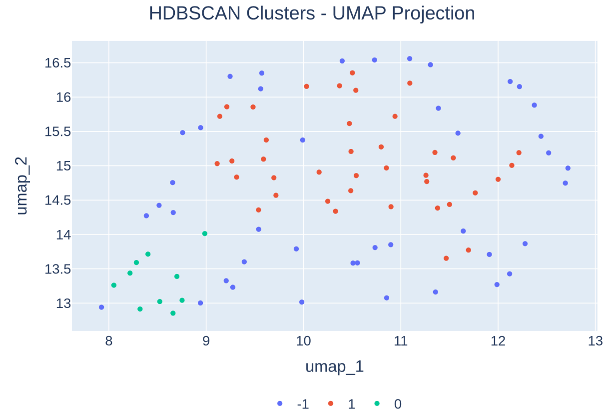

Fig. 5: HDBSCAN and UMAP Clusters

C. *Qualitative Analysis*

Besides the numerical results, listening to the songs helped to confirm that the computational timbre space captured meaningful similarities between some tracks.

One interesting case is the song "Desafinado". The version recorded by João Gilberto with the American saxophonist Stan Getz was grouped with Jazz songs—even though "Desafinado" is originally a Brazilian samba. When listening to this version, it sounds closer to Jazz: the harmony, the smooth rhythm, and the saxophone give it a different character. The model was able to reflect that, placing it next to other Jazz songs.

On the other hand, another version of "Desafinado", this time instrumental by Stan Getz and Charlie Byrd, was placed in a different cluster. This version sounds more like a traditional samba, with more rhythmic elements and a stronger percussive feel. It's curious that the version recorded with a Brazilian singer sounds more like Jazz, while the one recorded by two Americans sounds more like Samba—but this shows how the performance can change the perceived style.

Other Jazz songs were also grouped together, especially the ones with similar instruments and soft textures. In contrast, some Samba and Afrobeat tracks were spread out, possibly because these styles have more variety or the features used were not enough to capture their essence.

These examples suggest that even with a small dataset and simple features, the timbre space was able to capture some real musical similarities that make sense to the human ear.

D. *Conclusion*

This project investigated how computational models can represent the way humans perceive timbre. By applying dimensionality reduction and clustering techniques to a selected group of songs, visual representations of timbre space could be constructed. Among the tested methods, non-linear techniques like t-SNE and UMAP showed better results than PCA, especially when it comes to forming clear and meaningful groups.

The best cluster coherence was achieved using UMAP together with HDBSCAN, which produced compact and stylistically consistent clusters. One interesting example is how João Gilberto's "Desafinado" appeared near American Jazz tracks, showing that the model captured subtle similarities between genres from different cultures.

Although the results are promising, it's important to highlight some limitations. The dataset was intentionally small and focused on genres that are already perceptually similar, which makes it hard to generalize the findings. Also, the evaluation combined numerical metrics with subjective interpretation, which could be improved in future studies using listener tests or perceptual datasets.

Even with these limitations, the approach presented here shows potential for better understanding musical timbre and could support future work in music analysis, cultural studies, and recommendation systems.

References

[1] American National Standards Institute, *American National Standard: Psychoacoustical Terminology*. New York: ANSI, 1973.

[2] J. M. Grey, "Multidimensional perceptual scaling of musical timbres," *Journal of the Acoustical Society of America*, vol. 61, no. 5, pp. 1270–1277, 1977.

[3] D. M. Howard and J. A. S. Angus, *Acoustics and Psychoacoustics*, 5th ed. New York: Routledge, 2017.

[4] S. McAdams and E. Bigand, *Thinking in Sound: The Cognitive Psychology of Human Audition*. Oxford: Oxford University Press, 1993.

[5] B. C. J. Moore, *An Introduction to the Psychology of Hearing*, 6th ed. Leiden: Brill Academic Publishers, 2012.

[6] W. A. Sethares, *Tuning, Timbre, Spectrum, Scale*. New York: Springer, 2005.

[7] G. Von Bismarck, "Timbre of steady sounds: A factorial investigation of its verbal attributes," *Acustica*, vol. 30, no. 3, pp. 146–159, 1974.

[8] R. L. Pratt and P. E. Doak, "A subjective rating scale for timbre," *Journal of Sound and Vibration*, vol. 45, 1976.

[9] A. Zacharakis and J. Reiss, "An exploratory study of musical timbre through verbal descriptions," in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2011. [Online]. Available: https://www.eecs.qmul.ac.uk/~josh/documents/2011/ZacharakisReiss-2011-ISMIR.pdf

[10] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[11] R. J. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," *Pacific-Asia conference on knowledge discovery and data mining*, pp. 160–172, 2013.

[12] O. C. Jorge and H. S. Lopes, "Métodos de Aprendizado de Máquina Aplicados ao Reconhecimento Autoral e Temporal de Bandas de Rock and Roll," in *Anais do 14 Congresso Brasileiro de Inteligência Computacional*, B. J. T. Fernandes and A. {Pereira Júnior}, Eds., Curitiba, PR: ABRICOM, 2019, pp. 1–6.

[13] C. Guedes, K. Ganguli, C. Plachouras, S. Senturk, and A. J. Eisenberg, "Mapping Timbre Space in Regional Music Collections using Harmonic-Percussive Source Separation (HPSS) Decomposition," in *Proceedings of the 2nd International Conference on TImbre*, Sep. 2020.